



Monitoring and Alerting — TASCO Growth Platform

TASCO Insurance × VETC Growth Platform — Operations

iorta TechNXT

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Prepared by iorta TechNXT for TASCO Insurance. Confidential: for the use of TASCO Insurance and VETC only.

Contents

1. What the application exposes
2. Additional exporters (to configure)
3. Metric catalogue (from the code)
4. SLIs and SLOs
5. Golden signals and dashboards
6. Alert rules
7. Log-based alerts
8. Synthetic checks
9. Alert hygiene

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Monitoring and Alerting	1.0	07 October 2026	SRE (iorta TechNXT during hypercare, then TASCO IT operations)	Issued for submission

Item	Value
Stack (reference)	Prometheus (+ Alertmanager), Grafana, Loki (or the TASCO log platform), blackbox exporter, postgres_exporter / sql_exporter. Any Prometheus-compatible stack works
Related	Runbook and Support Guide (RB-xx) · Performance and Capacity Test Plan §4 SLOs · KI-14 (fixed), KI-17 (open)

1. What the application exposes

Endpoint	Content	Exposure
GET /metrics	Prometheus text format 0.0.4, per process : counters and histograms below, plus process_resident_memory_bytes and process_uptime_seconds	Protected by METRICS_TOKEN (KI-14 fixed): without Authorization: Bearer <token> the endpoint returns 401. Prometheus scrapes with authorization: {type: Bearer, credentials_file: ...}. The production Deployment mounts METRICS_TOKEN_FILE. It is open only if the token is unset, so always set it, block /metrics at the ingress, and let the NetworkPolicy allow only the Prometheus namespace
GET /health/live	{status:"ok"}	Liveness probe
GET /health/ready	{status, store, db}; 503 when not ready	Readiness probe; external synthetic
GET /api/ops/status	Circuit state per integration, active rule versions and checksums, eventBacklog by status, audit count	Staff with ops:read. Use for humans and dashboards' links, not as a scrape target
stdout JSON logs	request lines with requestId, method, route, status, ms; lifecycle and error events	Log pipeline (Loki/ELK)

Notes on the metrics implementation (src/shared/metrics.js):

- Counters are in-memory per pod and **reset on restart**. Always use `rate()` / `increase()` and `sum` across pods.
- The output has no `# TYPE` lines, so Prometheus ingests the series as untyped. `rate()` and `histogram_quantile()` still work on the `_bucket` series.
- Histogram buckets (seconds): 0.005, 0.01, 0.025, 0.05, 0.1, 0.25, 0.5, 1, 2.5, 5, 10, +Inf.
- `route` is the **route template** (e.g. `POST /api/customer/orders`, `GET /api/customers/:id`). Probes are `health`, the scrape itself is `metrics`, unmatched/static paths are `static`, and the spec is `openapi`, so cardinality is bounded (80 API routes + 4 labels).

2. Additional exporters (to configure)

Some signals the app does not export yet (KI-17). Provide them with a read-only SQL exporter against the **replica** (role `metrics_ro`), polled every 60 s. These series are prefixed `tasco_db_` to make clear they come from the database, not the app.

Series	Query (sketch)	Used by
tasco_db_domain_events{status}	SELECT status, count(*) FROM domain_events GROUP BY status	ALR-21, ALR-22
tasco_db_domain_events_oldest_pending_seconds	SELECT extract(epoch FROM now() - min(occurred_at)) FROM domain_events WHERE status='pending'	ALR-21
tasco_db_touchpoints_due_unexecuted	SELECT count(*) FROM touchpoints WHERE status='scheduled' AND due_date < CURRENT_DATE	ALR-34
tasco_db_handoffs_open_older_than_2h	SELECT count(*) FROM handoffs WHERE status='open' AND created_at < now() - interval '2 hours'	ALR-83
tasco_db_orders{status} (last 24 h)	SELECT status, count(*) FROM orders WHERE created_at > now()-interval '1 day' GROUP BY status	ALR-84
tasco_db_job_last_success_timestamp{kind}	SELECT kind, extract(epoch FROM max(started_at)) FROM job_runs WHERE data->>'status'='succeeded' GROUP BY kind	ALR-35
postgres_exporter standard	Connections, replication lag, locks, transaction wraparound, DB size	ALR-90-ALR-93

Only `reconciliation` and `retention` record `job_runs` today. CLI-run `journeys`, `recompute` and `relay` do not, so their freshness comes from logs (ALR-33) and the CronJob status (`kube_cronjob_status_last_successful_time` from kube-state-metrics).

3. Metric catalogue (from the code)

Metric	Type	Labels	Emitted by
http_requests_total	counter	route, method, status	app.js, every request
http_request_duration_seconds	histogram	route, method	app.js
integration_calls_total	counter	integration (vetc-wallet, tasco-core, voice-ai, app-push, zalo-zns, sms), result (ok, error)	resilience.js, every attempt including retries
integration_short_circuit_total	counter	integration	resilience.js, calls rejected while the circuit is open
events_published_total	counter	type (policy.issued, profiles.rebuilt, lead.recompute_requested, handoff.created, renewal.link_requested, claim.submitted, rules.activated)	outboxEventBus.publish
events_processed_total	counter	type, status (done, pending = will retry, dead_letter)	outboxEventBus.relay
messages_total	counter	channel (app_push, zalo_zns, sms), status (sent, failed, blocked), journey	journeyService.sendMessage
voice_calls_total	counter	outcome (hot_handoff, link_sent, already_renewed, opted_out, wrong_person, plate_mismatch, unverified, callback_later, not_interested)	voiceService.finalize
quotes_total	counter	channel (vetc_app, zalo, telesales, voice_bot, partner_api)	salesService.quote
orders_completed_total	counter	channel, journey	salesService.purchase
orders_compensation_total	counter	status (issuance_failed_refunded, compensation_failed)	salesService.purchase, saga compensation after an issuance failure
process_resident_memory_bytes	gauge	—	metrics.render
process_uptime_seconds	gauge	—	metrics.render

4. SLIs and SLOs

SLO ID	User journey	SLI (Prometheus)	Objective (28-day rolling)
SLO-01	Customer purchase path availability	Non-5xx ratio over route=~"POST /api/customer/(quotes orders) POST /api/customer/session GET /api/customer/home"	99.9 % (40 min error budget / 28 d)
SLO-02	Partner API availability	Non-5xx ratio over route=~"(POST GET) /api/partner/v1/.*)"	99.9 %
SLO-03	Staff console availability	Non-5xx ratio over all other /api/* routes	99.5 %
SLO-04	Certificate verification availability	Non-5xx ratio over GET /api/public/certificates/:certNo + synthetic SYN-02	99.95 %
SLO-05	Latency: customer reads	p95 of GET /api/customer/home	≤ 300 ms for 99 % of 5-min windows
SLO-06	Latency: quotes	p95 of quote routes	≤ 500 ms
SLO-07	Latency: orders	p95 of order routes	≤ 2.5 s
SLO-08	Latency: staff reads	p95 of GET /api/leads, GET /api/customers/:id, GET /api/handoffs	≤ 500 ms
SLO-09	Journey freshness	Daily journey run completes by 12:00 ICT with backlog tasco_db_touchpoints_due_unexecuted = 0	99 % of days
SLO-10	Event freshness	Oldest pending event < 5 min	99 % of the time

Recording rules:

```

groups:
- name: tasco-gp-recording
  rules:
  - record: route:http_requests:rate5m
    expr: sum by (route, status) (rate(http_requests_total[5m]))
  - record: slo:customer_purchase:error_ratio5m
    expr: |
      sum(rate(http_requests_total{route=~"POST /api/customer/(quotes|orders)|POST /api/customer/session|GET
/api/customer/home",status=~"5.."}[5m]))
      /
      sum(rate(http_requests_total{route=~"POST /api/customer/(quotes|orders)|POST /api/customer/session|GET
/api/customer/home"}[5m]))
  - record: slo:customer_purchase:error_ratio1h
    expr: |
      sum(rate(http_requests_total{route=~"POST /api/customer/(quotes|orders)|POST /api/customer/session|GET
/api/customer/home",status=~"5.."}[1h]))
      /
      sum(rate(http_requests_total{route=~"POST /api/customer/(quotes|orders)|POST /api/customer/session|GET
/api/customer/home"}[1h]))
  - record: route:http_request_duration_seconds:p95_5m
    expr: histogram_quantile(0.95, sum by (le, route)
(rate(http_request_duration_seconds_bucket{route!~"health|metrics|static|openapi"}[5m])))
  - record: integration:error_ratio5m
    expr: |
      sum by (integration) (rate(integration_calls_total{result="error"}[5m]))
      / sum by (integration) (rate(integration_calls_total[5m]))

```

5. Golden signals and dashboards

Dashboard	Panels	Audience
D1 Service overview	RPS by route group; 4xx/5xx ratio; p50/p95/p99 by route (route:http_request_duration_seconds:p95_5m); 429 rate; pods ready; memory per pod; restarts; SLO burn (SLO-01..04)	On-call
D2 Integrations	Per integration: call rate, error ratio, short-circuits, circuit state (from integration_short_circuit_total > 0 plus a link to /api/ops/status), retries (calls ÷ requests)	On-call, vendor managers
D3 Events and jobs	events_published_total vs events_processed_total{status} by type; tasco_db_domain_events{status}; oldest pending; dead letters; CronJob last success; journey run summary (log-derived)	On-call, L3
D4 Business funnel	quotes_total and orders_completed_total by channel and journey; quote → order conversion; messages by channel and status; voice outcomes; open handoffs > 2 h; GWP (from the /api/dashboard/overview export, daily)	Business owner, campaign managers
D5 Compliance	messages_total{status="blocked"}; marketing messages outside the window (SQL, should be 0); opt-outs (voice_calls_total{outcome="opted_out"}); DNC contacts (should be 0); audit chain verify status; rule activations timeline	Compliance, DPO
D6 Security	Login 401/423/429 rates; MFA failures (audit); 403 rate by route; partner 401 rate; WAF blocks	Security
D7 Database	Connections vs max, CPU, IOPS, replication lag, slow statements, table sizes (domain_events, audit_log, messages), advisory-lock waits	DBA

Golden signals mapping:

Signal	Metric
Latency	http_request_duration_seconds (by route), ms in logs
Traffic	rate(http_requests_total[5m]), quotes_total, orders_completed_total
Errors	http_requests_total{status=~"5.."}, integration_calls_total{result="error"}, events_processed_total{status="dead_letter"}, messages_total{status="failed"}
Saturation	process_resident_memory_bytes, container CPU throttling, DB connections, tasco_db_domain_events{status="pending"}

6. Alert rules

severity: `page` pages on-call 24×7; severity: `ticket` creates a ticket and posts to the channel during business hours. Every alert carries a `runbook_url` to the Runbook and Support Guide and names the RB-xx procedure.

6.1 Platform

ID	Name	Expression	For	Severity	Runbook
ALR-01	ReadinessFailing	probe_success{job="blackbox",instance=~".*/health/ready"} == 0	2m	page	RB-01
ALR-02	HighErrorRate	sum(rate(http_requests_total{status=~"5..",route!~"health metrics static"}[5m])) / sum(rate(http_requests_total{route!~"health metrics static"}[5m])) > 0.02	5m	page	RB-01 / RB-12

ID	Name	Expression	For	Severity	Runbook
ALR-03	LatencySLOBreach	route:http_request_duration_seconds:p95_5m{route=~"GET /api/customer/home"} > 0.3 or route:http_request_duration_seconds:p95_5m{route=~"POST /api/(customer/)?quotes POST /api/partner/v1/quotes"} > 0.5 or route:http_request_duration_seconds:p95_5m{route=~"POST /api/(customer/ partner/v1/)?orders"} > 2.5	10m	ticket (page if > 2x threshold)	RB-12
ALR-04	PurchaseSLOFastBurn	slo:customer_purchase:error_ratio5m > (14.4 * 0.001) and slo:customer_purchase:error_ratio1h > (14.4 * 0.001)	2m	page	RB-02/03
ALR-05	PurchaseSLOSlowBurn	slo:customer_purchase:error_ratio1h > (6 * 0.001)	30m	ticket	—
ALR-06	InternalErrors	sum(increase(http_requests_total{status="500"}[10m])) > 5 (500 = INTERNAL_ERROR; 503 is upstream)	—	ticket	Logs unhandled error
ALR-07	RateLimitSurge	sum(rate(http_requests_total{status="429"}[5m])) > 5	10m	ticket	RB-09 / RB-14
ALR-08	NoTraffic	sum(rate(http_requests_total{route!~"health metrics static"}[15m])) == 0 and on() (hour() ≥ 1 and hour() < 13) (08:00-20:00 ICT)	15m	page	RB-01
ALR-09	ScrapeDown	up{job="tasco-growth-api"} == 0	5m	ticket	—

6.2 Integrations

ID	Name	Expression	For	Severity	Runbook
ALR-10	CircuitOpen	sum by (integration) (increase(integration_short_circuit_total[5m])) > 0	1m	page for vetco-wallet, tasco-core; ticket for others	RB-02 – RB-05
ALR-11	IntegrationErrorRatio	integration:error_ratio5m > 0.2 and sum by (integration) (rate(integration_calls_total[5m])) > 0.1	5m	ticket	RB-02 – RB-05
ALR-12	IntegrationRetryInflation	sum by (integration) (rate(integration_calls_total[10m])) / on() group_left sum(rate(http_requests_total{route=~"POST /api/(customer/ partner/v1/)?orders"}[10m])) > 3 (wallet: > 1 call per order means retries)	15m	ticket	RB-02

6.3 Events and jobs

ID	Name	Expression	For	Severity	Runbook
ALR-20	DeadLetterEvents	sum by (type) (increase(events_processed_total{status="dead_letter"}[15m])) > 0	—	page (business hours) / ticket (night)	RB-06
ALR-21	OutboxBacklog	tasco_db_domain_events{status="pending"} > 10000 or tasco_db_domain_events_oldest_pending_seconds > 300	10m	ticket (page if > 30 min)	RB-06
ALR-22	EventsStuckProcessing	tasco_db_domain_events{status="processing"} > 0 sustained. The Postgres relay re-claims after 5 min, so persistence means no relay is running	15m	ticket	RB-06
ALR-23	EventRetryChurn	sum by (type) (rate(events_processed_total{status="pending"}[10m])) > 0.5	15m	ticket	RB-06
ALR-24	BacklogGrowth (app-only fallback)	sum(increase(events_published_total[30m])) - sum(increase(events_processed_total{status="done"}[30m])) > 5000	30m	ticket	RB-06
ALR-33	JourneySkipSpike (log-based)	Loki ruler: max_over_time({app="tasco-growth-api" json msg="journey run complete" unwrap skipped [1h]} / max_over_time({app="tasco-growth-api" json msg="journey run complete" unwrap due [1h]} > 0.3	—	ticket	RB-07
ALR-34	JourneyBacklog	tasco_db_touchpoints_due_unexecuted > 0 and on() hour() ≥ 5 (after 12:00 ICT)	30m	ticket	RB-07
ALR-35	JobNotRun	time() - kube_cronjob_status_last_successful_time{cronjob=~"tasco-growth-(journeys recompute reconcile retention)"} > 26*3600; for tasco-growth-relay: > 900 s	—	ticket	SOP-09
ALR-36	ReconciliationMismatch	Log/SQL: last reconciliation job result mismatches > 0 (from GET /api/ops/jobs, or tasco_db query on job_runs.data→'result'→'mismatches')	—	ticket (page if > 10)	RB-11

6.4 Security and audit

ID	Name	Expression	For	Severity	Runbook
ALR-40	AuditChainBroken	Synthetic SYN-07 result ok == false (exported as probe_success{probe="audit_verify"} == 0)	—	page	RB-08
ALR-50	LoginFailureSpike	sum(increase(http_requests_total{route="POST /api/auth/login", status="401"}[5m])) > 30	—	ticket (page if > 200)	RB-09
ALR-51	AccountLockouts	sum(increase(http_requests_total{route="POST /api/auth/login", status="423"}[15m])) > 5	—	ticket	RB-09
ALR-52	LoginRateLimited	sum(increase(http_requests_total{route=~"POST /api/auth/(login mfa)", status="429"}[5m])) > 20	—	ticket	RB-09

ID	Name	Expression	For	Severity	Runbook
ALR-53	MfaFailures (log/audit)	Audit auth.mfa_failed count > 20 in 15 min (SQL exporter on audit_log.action)	—	ticket	RB-09
ALR-54	PartnerAuthFailures	sum(increase(http_requests_total{route=~".*/api/partner/v1/.*",status="401"}[10m])) > 50	—	ticket	RB-15
ALR-55	ForbiddenSpike	sum by (route) (increase(http_requests_total{status="403"}[10m])) > 50 (possible IDOR probing)	—	ticket	Security
ALR-56	DemoModeInProduction	Synthetic GET /api/meta body demoMode:true, or GET /api/demo/totp/admin ≠ 404 (defence in depth: production refuses to start with DEMO_MODE=true)	—	page	Configuration
ALR-57	InMemoryStoreInProduction	Synthetic GET /health/ready body store ≠ "postgres" (defence in depth: production refuses to start without DATABASE_URL, KI-28 fixed)	—	page	RB-01

6.5 Resources

ID	Name	Expression	For	Severity	Runbook
ALR-60	MemoryHigh	max(process_resident_memory_bytes) > 0.8 * 1.5 * 1024^3 (adjust to the container limit)	15m	ticket	RB-13
ALR-61	FrequentRestarts	sum(resets(process_uptime_seconds[1h])) > 3	—	ticket	RB-13
ALR-62	CPUThrottling	sum(rate(container_cpu_cfs_throttled_periods_total{container="api"}[5m])) / sum(rate(container_cpu_cfs_periods_total{container="api"}[5m])) > 0.25	15m	ticket	RB-12

6.6 Business KPIs

ID	Name	Expression	For	Severity	Notes
ALR-30	CopyGuardBlocked	sum(increase(messages_total{status="blocked"}[1h])) > 0	—	page Compliance + L2	An active template contains banned wording, which validation should prevent: investigate the rule set (RB-10)
ALR-31	MessageFailureRatio	sum by (channel) (increase(messages_total{status="failed"}[30m])) / sum by (channel) (increase(messages_total[30m])) > 0.1	—	ticket	RB-04
ALR-80	NoOrdersInBusinessHours	sum(increase(orders_completed_total[1h])) = 0 and on() (hour() ≥ 2 and hour() < 13)	—	page (after pilot full volume)	09:00-20:00 ICT
ALR-81	ConversionDrop	sum(increase(orders_completed_total[6h])) / sum(increase(quotes_total[6h])) < 0.5 * (sum(increase(orders_completed_total[6h] offset 1w)) / sum(increase(quotes_total[6h] offset 1w)))	—	ticket	Week-over-week
ALR-82	QuotesDropByChannel	sum by (channel) (increase(quotes_total[1h])) < 0.3 * sum by (channel) (increase(quotes_total[1h] offset 1w))	—	ticket	Channel outage
ALR-83	HotHandoffsUnworked	tasco_db_handoffs_open_older_than_2h > 20 and on() (hour() ≥ 1 and hour() < 11)	30m	ticket to the telesales supervisor	KPI: first contact ≤ 2 business hours
ALR-84	IssuanceFailures	sum by (status) (increase(orders_compensation_total[15m])) > 0 (app metric), or tasco_db_orders{status=~"issuance_failed_refunded compensation_failed"} > 0	—	page (compensation_failed always pages)	RB-11
ALR-85	OptOutSpike	sum(increase(voice_calls_total{outcome="opted_out"}[2h])) / sum(increase(voice_calls_total[2h])) > 0.1	—	ticket to Compliance	Script or targeting problem
ALR-86	PlateMismatchSpike	sum(increase(voice_calls_total{outcome=~"plate_mismatch unverified"}[2h])) / sum(increase(voice_calls_total[2h])) > 0.15	—	ticket	Data quality, or ASR quality (spoken-plate parsing fixed under KI-25)

6.7 Database

ID	Name	Expression (postgres_exporter)	For	Severity
ALR-90	DBConnectionsHigh	sum(pg_stat_activity_count) / max(pg_settings_max_connections) > 0.8	5m	ticket
ALR-91	DBReplicationLag	pg_replication_lag_seconds > 30	5m	page (RPO at risk)
ALR-92	DBDiskHigh	DB volume used > 70 % (provider metric)	15m	ticket
ALR-93	BackupFailed	Provider backup/PITR status ≠ success in 24 h	—	page

Example Alertmanager rule file entry:

```

- alert: CircuitOpen
  expr: sum by (integration) (increase(integration_short_circuit_total[5m])) > 0
  for: 1m
  labels:
    severity: '{{ if or (eq $labels.integration "vetc-wallet") (eq $labels.integration "tasco-core") }}page{{ else }}ticket{{
end }}'

annotations:
  summary: "Circuit open for {{ $labels.integration }}"
  runbook_url: "<docs-portal>/operations/runbook-and-support-guide.pdf"
  runbook_section: "RB-02 to RB-05"

```

(Alertmanager cannot template labels in the rule itself on every version. If yours cannot, split the rule into a `page` rule for `integration=~"vetc-wallet|tasco-core"` and a `ticket` rule for the rest.)

7. Log-based alerts

ID	Query (LogQL)	Threshold	Severity
LOG-01	<code>sum(count_over_time({app="tasco-growth-api"} json msg="unhandled error" [10m]))</code>	> 5	ticket
LOG-02	<code>count_over_time({app="tasco-growth-api"} json msg="circuit opened" [5m])</code>	> 0	follows ALR-10
LOG-03	<code>count_over_time({app="tasco-growth-api"} json msg="job failed" [1h])</code>	> 0	ticket
LOG-04	<code>count_over_time({app="tasco-growth-api"} json msg="pg pool error" [5m])</code>	> 3	page
LOG-05	<code>count_over_time({app="tasco-growth-api"} json level="warn" ~ "not set – using an ephemeral"[1h])</code>	> 0 in PROD	page (missing secret in a non-production config that reached prod)
LOG-06	Sign-in failures: <code>sum(count_over_time({app="tasco-growth-api"} json route="POST /api/auth/login" status="401" [5m]))</code> . The audit action <code>auth.login_failed</code> is in the DB, not the logs; use the HTTP 401 on the login route as the log proxy	> 30	ticket
LOG-07	<code>{app="tasco-growth-api"} ~ "(?i)(\\b0[35789]\\d{8}\\b)"</code> (unredacted VN mobile pattern in logs)	> 0	ticket to Security (redaction gap)

8. Synthetic checks

ID	Check	Frequency / location	Assertion	Alert
SYN-01	GET /health/ready through the public ingress	1 min, 2 locations (HN, HCM)	200, db:true, store:"postgres"	ALR-01, ALR-57
SYN-02	GET /api/public/certificates/<synthetic cert> (a long-dated internal test policy agreed with TASCO core)	5 min	valid:true, masked plate	ALR-71
SYN-03	Staff login with a no-MFA synthetic user (role executive, read-only) → GET /api/auth/me	5 min	200 within 1 s	ticket
SYN-04	Customer flow (session → home → quote) in PREPROD only . In PROD a quote creates real records, so it runs read-only (session + home) for a synthetic test profile excluded from KPIs	10 min	200s within SLO	ticket
SYN-05	GET /api/meta	5 min	demoMode:false	ALR-56
SYN-06	GET /api/demo/totp/admin	1 h	404	ALR-56
SYN-07	GET /api/audit/verify with an auditor service account	1 h, off-peak heavy (KI-12)	ok:true	ALR-40
SYN-08	TLS certificate expiry for the public hostnames	Daily	> 21 days	ticket
SYN-09	/verify/<certNo> page renders (HTML 200)	15 min	200, contains the cert number	ALR-71
ALR-71	Certificate verification synthetic failing	SYN-02 or SYN-09 failing for 10 min	—	page (police/inspection depend on it)

9. Alert hygiene

- Every alert has an owner, a runbook link and a tested expression (unit-tested with `promtool test rules`).
- Silences must have an expiry and a ticket ID.
- Monthly review: alerts that fired with no action are tuned or removed; incidents with no alert get a new rule.
- Thresholds marked for the pilot are re-baselined after pilot week 2 and before W1 (volumes ×20–30).